



Namal University Mianwali

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Vehicle Transport Management System

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CHAPTER 1 PROJECT OVERVIEW

1.1 INTRODUCTION

Efficient vehicle transportation is essential for businesses that handle vehicle export services. Delivering on time, keeping accurate records, and coordinating well between departments are important factors for smooth operations and customer's satisfaction. However, many companies still use manual methods like record books or simple spreadsheets to manage their transportation tasks. These old-fashioned ways can lead to lost data, repeated records, delays in processing orders, and problems with tracking shipments. To solve these issues, we propose a Vehicle Transport Management System (VTMS) that brings in a computerized, database driven approach to managing vehicle transportation. This system will keep and manage all operational data in a structured database, making it easy to store, retrieve, and handle transportation related information. The VTMS will offer several important features like selecting vehicle types, placing orders, tracking shipments and processing payments. It will also use Role-Based Access Control (RBAC) to ensure that users can access the system based on their roles. The main users will include vehicle buyers, administrators, transport managers, and finance managers. By using a centralized database with MySQL, the system will improve data accuracy, cut down on duplicate records, and provide secure management of transportation information. Additionally, it will give real-time updates on shipment status through unique tracking IDs and automatic notifications. Overall, the Vehicle Transport Management System will offer a reliable, efficient, and scalable way to manage vehicle transportation operations in today's digital world.

1.2 PROBLEM STATEMENT

Vehicle buyers in Pakistan often face difficulties arranging reliable and timely transportation for their purchased vehicles. Existing transport services do not have a centralized system to manage shipments efficiently. This lack of coordination causes delays, inaccurate fee calculations, and limited transparency. Customers cannot easily track their vehicles in real time or verify deliveries securely. Payments and refunds are often unclear or handled inconsistently. Transportation companies also face challenges. They struggle to manage available trucks and

drivers. Coordinating shipment assignments and maintaining accurate records of vehicles, payments, and customer information is difficult. These issues lead to missed shipments, underused resources, and low customer satisfaction. The Vehicle Transport Management System (VTMS) will solve these problems. The system allows customers to book shipments, calculate fees automatically, and make payments securely. It verifies deliveries using OTP and tracks shipments in real time. The system also manages trucks and drivers efficiently. It stores accurate data and provides clear records. VTMS improves transparency, reliability, and the overall efficiency of domestic vehicle transportation services.

1.3 PROJECT OBJECTIVES

The primary objective of this project is to design and implement a database-driven Vehicle Transport Management System (VTMS) that efficiently manages vehicle transportation operations by storing, organizing, and processing all data related to vehicle orders, customers, shipments, and payments. The system replaces traditional manual record-keeping with a centralized MySQL database that ensures accurate tracking of all transportation activities, improves data management, and enhances coordination among all stakeholders in the vehicle transport process.

The specific objectives of the VTMS are as follows:

- **Enable Online Shipment Booking:** Allow customers to book vehicle shipments without prior registration. A unique CustomerID is auto-generated from the customer's phone number or email at the time of booking. Booking details including VehicleID, Shipment-Type, pickup and delivery locations, and booking date are stored securely in the Shipment table of the database
- **Calculate Shipment Fees Automatically:** Determine accurate fees based on vehicle category, shipment type, and distance using Google Maps API. Store the calculated fee in the database and display it transparently to the customer.
- **Handle Payments Efficiently:** Support advance and cash-on-delivery payments. Record all payment transactions in the database, track their status, and enforce the refund policy at the booking stage.
- **Verify Customers and Deliveries via OTP:** Generate unique OTPs at the time of booking and at the time of delivery confirmation. Store OTP records in the OTP table with fields for OTPID, ShipmentID, CustomerID, OTP_Value, Generated_Time, Expiry_Time, Verification_Status, Purpose, and VerifiedAt. Prevent confirmation if the OTP is expired or invalid.

- **Track Shipments:** Maintain a complete Shipment_Status_Log for every shipment, recording Old_Status, New_Status, ChangedBy, Changed_Time, Notes, and Current_Location at each status transition (Booked → Scheduled → In Transit → Delivered). Provide both customers and transport managers with access to current shipment status and ETA.
- **Manage Trucks, Drivers, and Driver Assignments:** Maintain up-to-date Truck and Driver records in the database. Use the Driver_Assignment table (with fields AssignmentID, DriverID, TruckID, Assignment_Date, End_Date, and Status) to manage which driver and truck are assigned to each shipment. Allow transport managers to update statuses and handle unavailability by marking shipments as Pending
- **Ensure Data Accuracy and Security:** Store and manage all customer, shipment, vehicle, payment, OTP, and notification data reliably in MySQL. Implement Role-Based Access Control (RBAC) via the User table, restricting system access to authorized roles: Transport Manager, Finance Manager, Driver, and Admin.
- **Support Operational Transparency:** Generate unique TrackingIDs for every shipment. Record all shipment status changes in the Shipment_Status_Log and all notifications in the Notification table. Produce operational and financial reports that support monitoring, auditing, and decision-making across the organization.

GitHub Repository: <https://github.com/MominaUmar74/VTMS.git>

CHAPTER 2 DETAILED DATABASE DESIGN

2.1 ENTITY

The following table identifies and defines all entities in the VTMS database. Each entity represents a distinct real world concept central to the vehicle transportation workflow. Associative entities arising from many-to-many relationships (such as Driver_Assignment) are included here as they appear as named tables in the ERD.

Sr. No	Entity Name	Description
01	Customer	An individual who books a vehicle shipment. CustomerID is auto-generated from phone or email; no pre-registration is required.
02	User	An internal staff member with a defined system role (Transport Manager, Finance Manager, Driver, or Admin) who accesses the VTMS backend.
03	Driver	An employee responsible for operating an assigned truck and delivering vehicles. Extends User with license and status details.
04	Vehicle	The vehicle to be transported. Linked to a Vehicle_Category and contains make, model, year, color, and registration details.
05	Vehicle_Category	A lookup table defining the five vehicle categories (Motorcycle, Sedan, SUV/MPV, Pickup, Luxury) along with their base rate per km and fee multiplier.
06	Shipment	A vehicle transport order that captures pickup/delivery locations, booking date, distance, fee, shipment status, delivery date, payment status, TrackingID, and AssignmentID.
07	Truck	A company-owned transport vehicle used to carry customer vehicles. Tracks truck type, capacity, and current operational status.

08	Driver_Assignment	An associative entity recording which driver and truck are assigned to a shipment, including assignment date, end date, and assignment status.
09	Shipment_Status_Log	An audit trail table that records every status change for a shipment, including old status, new status, who changed it, timestamp, notes, and current location.
10	Payment	Records all financial transactions linked to a shipment, including advance and COD payments, refund status, and which staff member processed the transaction.
11	OTP	Stores one-time passwords generated for booking confirmation and delivery verification, including expiry time and verification result.
12	Notification	Records all automated messages sent to customers or staff regarding booking, payment, OTP verification, or shipment status updates.

Table 2.1: VTMS Entity Definitions

2.2 DATA DICTIONARY

The data dictionary below defines every attribute of each entity identified in the ERD, including data type, constraint, and description. Foreign Keys (FK) are noted in the attribute name and description.

2.2.1 Customer

Stores information about the vehicle owner who places a shipment booking. No pre-registration is required; the CustomerID is generated automatically from the customer's contact details.

Sr. No	Name	Data Type	Constraint	Description
01	Customer_ID (PK)	VARCHAR(20)	NOT NULL, UNIQUE	Auto-generated unique identifier derived from the customer's phone number or email at booking time.
02	Name	VARCHAR(100)	NOT NULL	Full name of the customer.

03	Phone	VARCHAR(15)	NOT NULL, UNIQUE	Customer's primary contact phone number; used for OTP delivery.
04	Email	VARCHAR(100)	UNIQUE, NULL	Customer's email address (optional).
05	Address	VARCHAR(255)	NULL	Customer's residential or business address for reference.

2.2.2 User

Represents all internal staff members of the transport company. The Role attribute enforces access control through RBAC. The Driver entity is a specialisation of User and shares the same UserID as its primary key

Sr. No	Name	Data Type	Constraint	Description
01	UserID (PK)	VARCHAR(15)	NOT NULL, UNIQUE	Unique identifier for each internal system user (staff member).
02	UserName	VARCHAR(100)	NOT NULL, UNIQUE	Login username of the staff member.
03	Password	VARCHAR(255)	NOT NULL	Hashed password stored securely (e.g. bcrypt). Never stored in plaintext.
04	Role	ENUM	NOT NULL	User role: Transport Manager, Finance Manager, Driver, or Admin.
05	Phone	VARCHAR(15)	NOT NULL, UNIQUE	Staff member's contact phone number.
06	Email	VARCHAR(100)	NOT NULL, UNIQUE	Official email address used for login and notifications.
07	Salary	DECIMAL(10,2)	NULL	Monthly salary of the staff member (used for HR reporting by Admin).
08	ReportsTo	VARCHAR(15)	NULL (FK UserID)	Self-referencing FK indicating the staff member's immediate supervisor.

09	Joining_Date	DATE	NOT NULL	Date on which the staff member joined the organization.
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2.2.3 Driver

A specialization of the User entity. Every Driver record references an existing User record where Role = Driver. The Driver entity adds license and status information specific to driving personnel.

Sr. No	Name	Data Type	Constraint	Description
01	UserID (PK/FK)	VARCHAR(15)	NOT NULL, UNIQUE	References the User table; the Driver is a specialized User with Role = Driver.
02	Licence_No	VARCHAR(30)	NOT NULL, UNIQUE	Official driving license number issued by the relevant authority.
03	Current_Status	ENUM	NOT NULL	Current operational status: Available, On Duty, or Off Duty.

2.2.4 Vehicle

Represents the vehicle to be transported. Each vehicle is linked to a Vehicle_Category that determines the fee calculation parameters. A vehicle can be associated with multiple shipments over its lifetime.

Sr. No	Name	Data Type	Constraint	Description
01	VehicleID (PK)	VARCHAR(20)	NOT NULL, UNIQUE	Unique identifier for the vehicle being transported.
02	Registration_N	VARCHAR(30)	NOT NULL, UNIQUE	Official vehicle registration (license plate) number.
03	Make	VARCHAR(50)	NOT NULL	Manufacturer of the vehicle (e.g. Toyota, Honda, Suzuki).
04	Model	VARCHAR(50)	NOT NULL	Model name of the vehicle (e.g. Corolla, Civic, Alto).

05	CategoryID (FK)	VARCHAR(10)	NOT NULL	References the Vehicle_Category table to determine the fee multiplier and base rate.
06	Year	YEAR	NOT NULL	Manufacturing year of the vehicle.
07	Color	VARCHAR(30)	NULL	Color of the vehicle for identification purposes.
08	Description	TEXT	NULL	Any additional description or notes about the vehicle's condition.
09	Record_CreatedAt	DATETIME	NOT NULL, DEFAULT NOW()	Timestamp recording when the vehicle record was created in the system.

2.2.5 Vehicle_Category

A lookup/reference table defining the five vehicle categories supported by the VTMS. The BaseRatePerKM and Multiplier values stored here are used directly in the automated fee calculation module.

Sr. No	Name	Data Type	Constraint	Description
01	CategoryID (PK)	VARCHAR(10)	NOT NULL, UNIQUE	Unique identifier for the vehicle category.
02	CategoryName	VARCHAR(50)	NOT NULL, UNIQUE	Name of the category: Motorcycle, Sedan, SUV/MPV, Pickup, or Luxury.
03	BaseRatePerKM	DECIMAL(8,2)	NOT NULL	Base transportation rate in PKR per kilometer for this category.
04	Multiplier	DECIMAL(4,2)	NOT NULL	Fee multiplier applied during calculation. Exclusive shipments use a higher multiplier than Shared.

2.2.6 Shipment

The central entity of the VTMS. A Shipment record encapsulates the complete lifecycle of a vehicle transport order from initial booking through driver assignment, in-transit tracking, and final delivery confirmation. The AssignmentID FK links the shipment to its assigned driver and truck.

Sr. No	Name	Data Type	Constraint	Description
01	Shipment_ID (PK)	VARCHAR(20)	NOT NULL, UNIQUE	Auto-generated unique shipment identifier used as the tracking reference.
02	CustomerID (FK)	VARCHAR(20)	NOT NULL	References the Customer who placed this shipment booking.
03	VehicleID (FK)	VARCHAR(20)	NOT NULL	References the Vehicle entity representing the vehicle being transported.
04	Pickup_Location	VARCHAR(255)	NOT NULL	Full address or city name of the vehicle pickup point.
05	Delivery_Location	VARCHAR(255)	NOT NULL	Full address or city name of the vehicle delivery destination.
06	Booking_Date	DATETIME	NOT NULL, DEFAULT NOW()	Timestamp when the shipment was booked by the customer.
07	Shipment_Status	ENUM	NOT NULL	Current status: Booked, Scheduled, In Transit, Delivered, or Pending.
08	Distance	DECIMAL(8,2)	NOT NULL	Distance in kilometers between pickup and delivery, calculated via Google Maps API.
09	Fee	DECIMAL(10,2)	NOT NULL	Total calculated shipment fee in PKR (based on category rate, multiplier, and distance).
10	AssignmentID (FK)	VARCHAR(20)	NULL	References the Driver_Assignment record linking the truck and driver to this shipment.
11	TrackingID	VARCHAR(30)	NOT NULL, UNIQUE	Unique customer-facing tracking code generated at booking.
12	Delivery_Date	DATETIME	NULL	Actual date and time of delivery, recorded when status changes to Delivered.

13	Payment_Status	ENUM	NOT NULL	Payment status for this shipment: Pending, Paid, or Refunded.
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2.2.7 Truck

Represents company-owned transport trucks. The Current_Status attribute is updated automatically when a truck is assigned to or released from a shipment, ensuring accurate availability tracking.

Sr. No	Name	Data Type	Constraint	Description
01	TruckID (PK)	VARCHAR(15)	NOT NULL, UNIQUE	Unique identifier for each company truck.
02	Truck_Type	VARCHAR(50)	NOT NULL	Type or model classification of the truck (e.g. flatbed, enclosed carrier).
03	Capacity	INT	NOT NULL	Maximum number of vehicles the truck can carry in a single shipment.
04	Current_Status	ENUM	NOT NULL	Operational status: Available, In Transit, or Maintenance.

2.2.8 Driver Assignment

An associative table that links a Driver and a Truck to a Shipment for the duration of a delivery. Each active shipment has at most one active assignment record. The AssignmentID from this table is stored as a FK in the Shipment table.

Sr. No	Name	Data Type	Constraint	Description
01	AssignmentID (PK)	VARCHAR(20)	NOT NULL, UNIQUE	Auto-generated unique identifier for each driver–truck assignment record.
02	DriverID (FK)	VARCHAR(15)	NOT NULL	References the Driver (via UserID) assigned to this shipment.
03	TruckID (FK)	VARCHAR(15)	NOT NULL	References the Truck assigned to this shipment.

04	Assignment_Date	DATETIME	NOT NULL	Date and time when the driver and truck were assigned to the shipment.
05	End_Date	DATETIME	NULL	Date and time when the assignment ended (truck and driver released).
06	Status	ENUM	NOT NULL	Current assignment status: Active, Completed, or Cancelled.

2.2.9 Shipment_Status_Log

An audit trail table that records every change in shipment status. This ensures full transparency and traceability for customers and management. The Current_Location field provides real-time location context at the time of each update.

Sr. No	Name	Data Type	Constraint	Description
01	LogID (PK)	INT	NOT NULL, AUTO_INCREMENT	Auto-incremented unique identifier for each status change log entry.
02	ShipmentID (FK)	VARCHAR(20)	NOT NULL	References the Shipment whose status changed.
03	Old_Status	ENUM	NOT NULL	The previous status of the shipment before this change.
04	New_Status	ENUM	NOT NULL	The updated status of the shipment after this change.
05	ChangedBy (FK)	VARCHAR(15)	NOT NULL	References the User (staff member) who made this status change.
06	Changed_Time	DATETIME	NOT NULL	Exact timestamp when the status change was recorded in the database.
07	Notes	TEXT	NULL	Optional remarks about the reason for the status change or relevant observations.
08	Current_Location	VARCHAR(255)	NULL	Geographic location of the shipment at the time of this status update.

2.2.10 Payment

Records all financial transactions linked to a shipment. The ProcessedBy FK links the payment to the Finance Manager who handled it. Refunds are tracked through the Refund_Status attribute and are only permitted before the shipment is Scheduled.

Sr. No	Name	Data Type	Constraint	Description
01	PaymentID (PK)	VARCHAR(20)	NOT NULL, UNIQUE	Unique identifier for each payment transaction record.
02	ShipmentID (FK)	VARCHAR(20)	NOT NULL	References the Shipment to which this payment belongs.
03	CustomerID (FK)	VARCHAR(20)	NOT NULL	References the Customer who made or owes this payment.
04	Payment_Date	DATETIME	NOT NULL	Date and time the payment transaction was recorded.
05	Payment_Status	ENUM	NOT NULL	Status of the payment: Pending, Completed, or Refunded.
06	Refund_Status	ENUM	NOT NULL	Refund state: None, Requested, or Refunded. Refunds are only permitted at the booking stage.
07	CustomerID (FK)	VARCHAR(20)	NOT NULL	Duplicate FK reference ensuring payment is always linked to the correct customer.
08	ProcessedBy (FK)	VARCHAR(15)	NOT NULL	References the User (Finance Manager) who verified or processed this payment.
09	Amount	DECIMAL(10,2)	NOT NULL	Total transaction amount in PKR.
10	Payment_Type	ENUM	NOT NULL	Advance (minimum 30% of total fee) or Cash-on-Delivery (COD).
11	Payment_Method	VARCHAR(50)	NULL	Gateway used for advance payments: Easypaisa or JazzCash.

2.2.11 OTP

Stores one-time passwords generated at the time of booking (Purpose = Booking Confirmation) and at the time of delivery (Purpose = Delivery Confirmation). An expired or invalid OTP prevents the confirmation from proceeding.

Sr. No	Name	Data Type	Constraint	Description
01	OTPID (PK)	INT	NOT NULL, AUTO_INCREMENT	Auto-incremented unique identifier for each OTP record.
02	ShipmentID (FK)	VARCHAR(20)	NOT NULL	References the Shipment for which this OTP was generated.
03	CustomerID (FK)	VARCHAR(20)	NOT NULL	References the Customer who receives the OTP.
04	OTP_Value	VARCHAR(6)	NOT NULL	The 6-digit one-time password sent to the customer.
05	Generated_Time	DATETIME	NOT NULL	Timestamp when the OTP was generated.
06	Expiry_Time	DATETIME	NOT NULL	Timestamp after which the OTP is invalid (typically 10 minutes after generation).
07	Verification_Status	ENUM	NOT NULL	Status: Pending, Verified, or Expired.
08	Purpose	ENUM	NOT NULL	Reason for the OTP: Booking Confirmation or Delivery Confirmation.
09	VerifiedAt	DATETIME	NULL	Timestamp recording the exact moment the OTP was successfully verified.

2.2.12 Notification

Records all automated notifications dispatched to customers and internal staff via the JazzSMS API. Notifications cover all key system events: booking confirmation, payment confirmation, OTP delivery, shipment status updates, and delivery confirmation.

Sr. No	Name	Data Type	Constraint	Description
01	NotificationID (PK)	INT	NOT NULL, AUTO_INCREMENT	Autoincremented unique identifier for each notification record.

02	CustomerID (FK)	VARCHAR(20)	NULL	References the Customer recipient, if the notification targets a customer.
03	UserID (FK)	VARCHAR(15)	NULL	References the User (staff) recipient, if the notification targets internal staff.
04	ShipmentID (FK)	VARCHAR(20)	NULL	References the related Shipment, if applicable.
05	Title	VARCHAR(150)	NOT NULL	Short heading or subject line of the notification message.
06	Message	TEXT	NOT NULL	Full content of the notification message sent to the recipient.
07	Type	ENUM	NOT NULL	Category: Booking, Payment, OTP, Status Update, or Delivery.
08	SentAt	DATETIME	NOT NULL	Timestamp recording when the notification was dispatched.
09	Status	ENUM	NOT NULL	Delivery status of the notification: Sent or Failed.

2.3 BUSINESS RULES

The following business rules define the conditions, constraints, and logic that govern how data is created, managed, and related within the VTMS. These rules ensure data consistency, integrity, and correct system behavior across all operational modules.

2.3.1 Customer and Booking Rules

1. A Customer can place one or more Shipments. Each Shipment belongs to exactly one Customer (1:M relationship).
2. A CustomerID is auto-generated from the customer's phone number or email at the time of booking. No pre-registration account is required.
3. Each Shipment must reference exactly one Vehicle, and that Vehicle must belong to a valid Vehicle_Category.
4. A Shipment booking cannot be confirmed without successful OTP verification (OTP Purpose = Booking Confirmation).

5. Each Shipment must have a valid pickup location and a delivery location that are different from each other.

2.3.2 Fee Calculation Rules

1. Shipment fee is calculated as: $\text{Fee} = \text{BaseRatePerKM} \times \text{Distance} \times \text{Multiplier}$, where BaseRatePerKM and Multiplier are retrieved from the Vehicle_Category table based on the vehicle's category.
2. Distance is calculated in real time using the Google Maps API at the time of booking and stored in the Shipment table.
3. The calculated fee must be displayed to the customer and confirmed before the booking is finalized in the database.
4. Exclusive shipment types apply a higher Multiplier than Shared shipment types, as configured in the Vehicle_Category table.

2.3.3 Payment Rules

1. Each Shipment must have exactly one Payment record. A Payment record is created when the customer selects a payment method at booking.
2. Payment_Type is either Advance (minimum 30
3. Refund_Status may be set to Requested or Refunded only when Shipment_Status is Booked. Once the shipment is Scheduled or In Transit, refund requests are automatically rejected.
4. Processed By field in the Payment table must reference a User with Role = Finance Manager.

2.3.4 OTP Verification Rules

1. An OTP (Purpose = Booking Confirmation) is generated immediately when a booking is submitted and sent to the customer via the JazzSMS API.
2. A second OTP (Purpose = Delivery Confirmation) is generated when the driver initiates delivery confirmation and sent to the customer.
3. An OTP expires after 10 minutes ($\text{Expiry_Time} = \text{Generated_Time} + 10 \text{ minutes}$). An expired OTP cannot be used for verification and a new OTP must be requested.

4. If OTP Verification_Status is not Verified at the time of delivery, the delivery cannot be marked as Delivered in the Shipment table.

2.3.5 Truck and Driver Assignment Rules

1. A Truck's Current_Status must be Available before it can be included in a new Driver_Assignment. A Truck with status in Transit or Maintenance cannot be assigned.
2. A Driver's Current_Status must be Available before they can be included in a new Driver_Assignment.
3. Only a User with Role Transport Manager can create or modify Driver_Assignment records.
4. If no available Truck or Driver exists at the time of scheduling, the Shipment_Status is set to Pending and the customer receives a Notification of the delay.
5. When an assignment is completed (End_Date is populated), both the Driver's Current_Status and the Truck's Current_Status must be reset to Available.

2.3.6 Shipment Status and Tracking Rules

1. A Shipment must progress through statuses strictly in sequence: Booked → Scheduled → In Transit → Delivered. No status may be skipped.
2. Every status change must be recorded as a new row in the Shipment_Status_Log table, capturing Old_Status, New_Status, ChangedBy, Changed_Time, Notes, and Current_Location.
3. Only a User with Role Driver may change Shipment_Status from In Transit to Delivered, and only for Shipments assigned to them.
4. Only a User with Role Transport Manager may change Shipment_Status from Booked to Scheduled or from Scheduled to In Transit.
5. The TrackingID generated at booking must be unique across all Shipment records and is used by the customer to monitor real-time progress.

2.3.7 Role-Based Access Control (RBAC) Rules

1. Admin: Full read/write access to all tables User, Customer, Vehicle, Vehicle_Category, Shipment, Truck, Driver, Driver_Assignment, Shipment_Status_Log, Payment, OTP, and Notification. Can configure system settings, fees, and refund rules.

2. Transport Manager: Full read/write access to Shipment, Truck, Driver, Driver_Assignment, Shipment_Status_Log, and Notification. Read-only access to Customer and Payment data. Cannot modify User accounts or Payment records.
3. Finance Manager: Read/write access to Payment records only. Read-only access to Shipment and Customer data. Cannot modify Trucks, Drivers, Assignments, or Shipment status.
4. Driver: Read access to their own assigned Shipment records only. Write access to update Shipment_Status (In Transit → Delivered) and OTP Verification_Status for their assigned shipments. Cannot access Payment records, other Driver records, or unassigned Shipments.
5. Customer: Read/write access to their own booking, Shipment, and Payment records. Can view TrackingID and current status but cannot modify Truck, Driver, or other Customer data.

2.4 ENTITY RELATIONSHIP DIAGRAM

The Entity Relationship Diagram (ERD) below was created using draw.io (diagrams.net) and illustrates all entities, attributes, primary keys, foreign keys, relationships, and cardinalities within the VTMS database. All entity names, attribute names, and relationship lines are taken directly from the submitted ERD file.

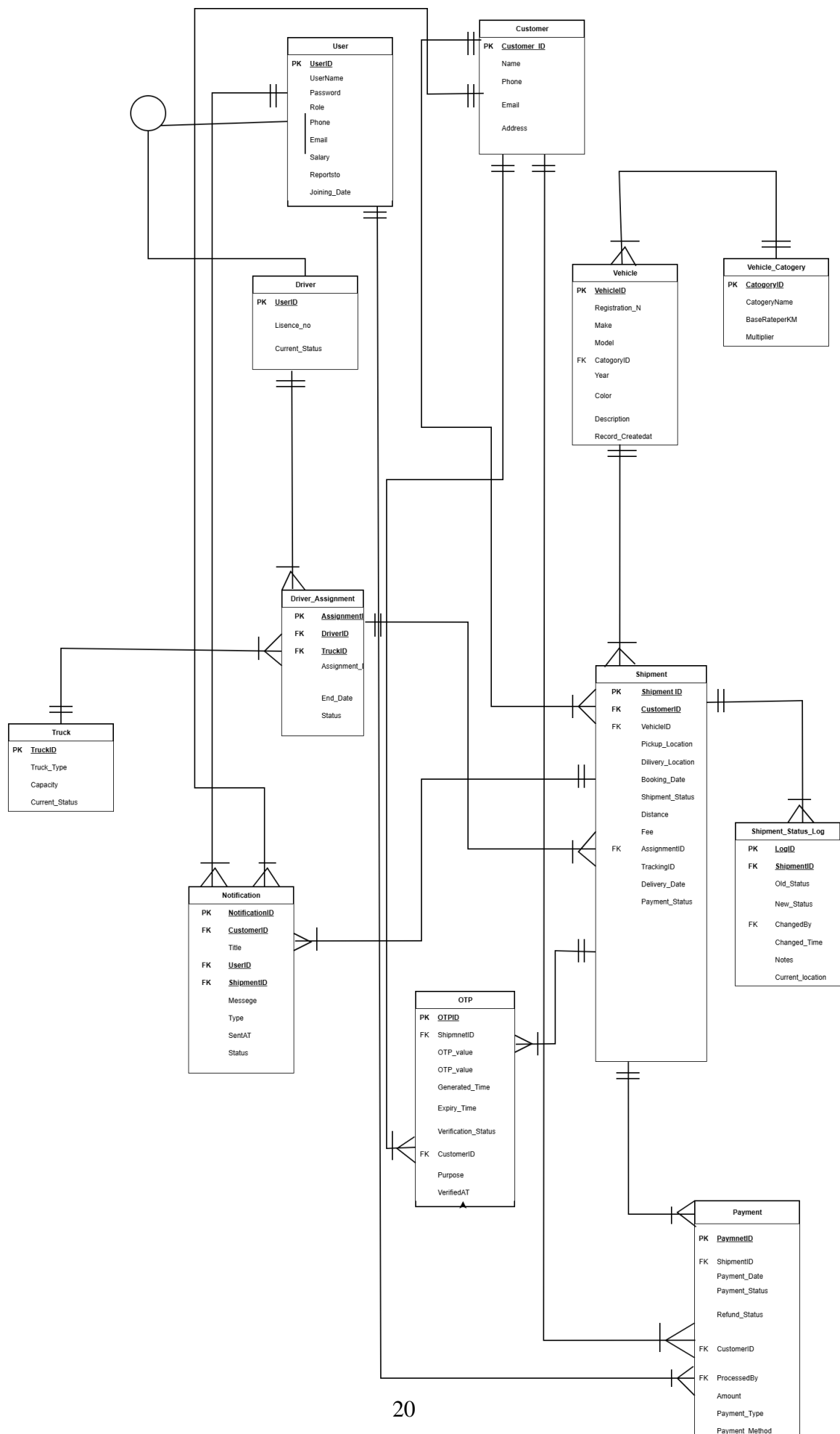


Figure 2.1: VTMS Entity Relationship Diagram (draw.io)

REFERENCES

1. CarCarrier.pk, "Vehicle Transport Services in Pakistan," [Online]. Available: <https://www.carcarrier.pk>. [Accessed: April 2026].
2. Google LLC, "Google Maps Platform — Distance Matrix and Directions API Documentation," [Online]. Available: <https://developers.google.com/maps>. [Accessed: April 2026].
3. Jazz Telecom Pvt. Ltd., "JazzSMS API Documentation for OTP and SMS Notifications," [Online]. Available: <https://www.jazz.com.pk/developer>. [Accessed: April 2026].
4. JazzCash / Easypaisa, "Developer Payment Gateway and Integration APIs," [Online]. Available: <https://www.jazzcash.com.pk/developer>. [Accessed: April 2026].
5. Oracle Corporation, "MySQL 8.0 Reference Manual," [Online]. Available: <https://dev.mysql.com/doc/refman/8.0/en>. [Accessed: April 2026].
6. OpenJS Foundation, "Node.js v20 Documentation," [Online]. Available: <https://nodejs.org/en/docs>. [Accessed: April 2026].
7. Meta Platforms Inc., "React 18 Documentation," [Online]. Available: <https://react.dev>. [Accessed: April 2026].
8. diagrams.net, "draw.io ERD Diagramming Tool," [Online]. Available: <https://app.diagrams.net>. [Accessed: April 2026].
9. VTMS GitHub Repository, [Online]. Available: <https://github.com/MominaUmar74/VTMS.git>. [Accessed: April 2026].
10. OpenAI, "ChatGPT AI Assistant," Conversation Reference: <https://chatgpt.com/c/69b25b22-37fc-83ab-bb9f-89a521259246>. [Accessed: March–April 2026].

APPENDIX: AI USAGE AND PROMPTS

This section documents all use of AI tools during the development of the VTMS project. The table below lists each prompt submitted to an AI assistant, along with the specific purpose for which it was used.

AI Tool Used: ChatGPT (OpenAI)

Reference Conversation: <https://chatgpt.com/c/69b25b22-37fc-83ab-bb9f-89a5212592>

Sr.	Prompt Used	Purpose
1	I am working on a database project for a Vehicle Transport Management System (VTMS). We have noticed that in Pakistan the manual record-keeping transport sector leads to tracking issues and payment transparency. Can you help me outline a proposal structure that covers everything from Problem Statement to Technology Stack?	Structuring the overall project proposal. Used to generate an initial framework covering all required sections.
2	For our VTMS, we decided on five vehicle categories: Motorcycle, Sedan, SUV/MPV, Pickup, and Luxury. Does this cover the standard needs for a domestic export service, and what specific data attributes should we store in a MySQL table to track these shipments effectively?	Validating vehicle categories and identifying appropriate MySQL table attributes for the Vehicle and Vehicle_Category entities.
3	Our project uses Node.js and MySQL. We want to automate fee calculation using the Google Maps API and send delivery verifications via JazzSMS. How do these two APIs typically interact with a backend to update a shipment status in real time?	Understanding the API integration pattern between the Google Maps API (distance/ETA), JazzSMS API (OTP/notifications), and the Node.js/MySQL backend.

4	I have drafted the conclusion for our VTMS project. Can you review it to ensure it clearly emphasises how the system increases efficiency and customer satisfaction through centralised records?	Reviewing and refining the project conclusion for clarity, completeness, and academic tone.
5	We are implementing Role-Based Access Control. Can you help us define the specific database permissions for a Transport Manager versus a Finance Manager to ensure data security?	Defining and validating RBAC rules for different user roles to ensure appropriate separation of data access and system permissions.

Table 2.14: AI Prompts Used and Their Purposes